

TELCONG NETPLAN

Network Planning & Regulatory Affairs

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Network Design Standards - North-Central Region

Coverage planning, capacity standards, and site acquisition standards

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Region	North-Central (FCT, Nasarawa, Niger, Plateau, Kogi, Kwara, Benue)
Linked confluence	pg_design_standards

1. Coverage planning standards

Coverage planning for the North-Central region is anchored in the following standards. RSRP target at cell edge: at least -98 dBm for 2G/3G/4G; at least -95 dBm for 5G. Cell-edge SINR target: at least 4 dB on the dominant cell. Coverage probability target: at least 95% area coverage at the cell-edge thresholds for outdoor scenarios, at least 90% for indoor scenarios (based on the typical Nigerian residential construction loss of 12 dB). Coverage planning shall use the Ericsson 9999 propagation model calibrated against the 2024-2025 measurement campaigns; alternative models may be used with documented justification.

2. Capacity standards

Capacity planning for the region anchors on a busy-hour PRB utilisation target of below 70% on average, with no sector exceeding 85% during the busy hour. Densification trigger: when measured 4-week average busy-hour PRB utilisation crosses 70% on a sector, the sector enters the densification or carrier-add candidate list within 30 days. Capacity headroom for new site launches: at least 40% of the predicted busy-hour traffic must be growth headroom.

3. Site acquisition standards

Site acquisition shall follow the standard 4-stage process: (1) site identification and candidate ranking; (2) traditional leader and community engagement; (3) environmental and regulatory approval; (4) lease negotiation and commercial close. Each stage shall be documented with named owner and timeline. Standard cycle time: 90 days from candidate identification to lease execution, target; 120 days, upper bound; escalate beyond 120 days. The Nasarawa Phase 1 deployment is the live operational test of this standard.

4. Community engagement standards

Community engagement for new sites shall include, as a minimum: a formal letter to the traditional authority (chief, district head, or equivalent) before the first site visit; an in-person site visit with the traditional authority or their nominee present; a town hall briefing for residents within 300 metres of the proposed site; and a documented consent record signed by the traditional authority before site preparation begins. Objection threads are documented in the community engagement playbook (linked separately).

5. Environmental approval standards

Environmental approval shall be obtained from the Federal Ministry of Environment and the relevant State Environmental Protection Agency before site preparation begins. EMF compliance shall be assessed against the methodology documented in the EMF Compliance Methodology document (linked separately). Where the site is within 500 metres of a sensitive receptor (school, hospital, place of worship, residence with children under 12), the EMF assessment shall apply the additional 4 dB margin for sensitive populations.

6. Vendor coordination standards

All vendor equipment shall be procured via a duly-issued Purchase Order with the equipment vendor (Tachyon Networks Nigeria Ltd for the current pilot programme). Acceptance testing shall follow the vendor acceptance test plan (linked separately). NCC type-approval status shall be verified at the time of PO issuance and re-verified at the time of delivery. Equipment without current type-approval shall not be installed under any circumstances.

7. Regulatory compliance standards

All regulatory filings shall be made via the NCC portal under the named Authorised Signatory (currently Sheila Stokes for the Network Planning track). Filing submissions shall be cross-checked by the Regulatory Liaison (Fatima Musa) before submission. The 60-day Commission determination period shall be tracked against an expected response date, with escalation to the Director of Network Operations if no determination is received by day 55.

8. Documentation standards

All design decisions shall be documented on the engineering wiki (confluence FCBD or ENG space as appropriate). Document versions shall be incremented on material change. Filing positions submitted to the NCC shall align with the most recent version of the corresponding technical position document; alignment shall be cross-checked at the time of filing submission and documented in a consistency check page (e.g. pg_filing_position_consistency_check). Document owners are responsible for maintaining version history and notifying downstream dependants of material changes.